

Unit 21

ORGANIC SYNTHESIS

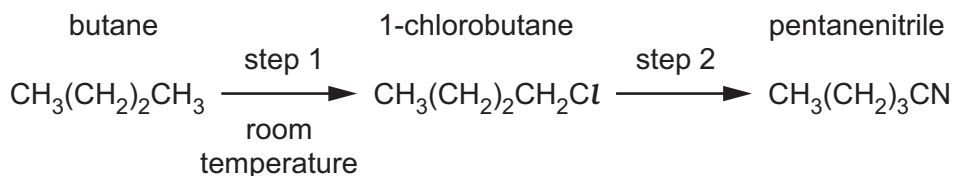
27 items · 112 marks

Syllabus sections in this unit

21.1 Organic synthesis

Unit 21: Organic synthesis**1. M/J 26/P24/Q4/d(i)**

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

**Figure 4.3**

(i) Suggest the reagent and conditions for each step in Figure 4.3.

step 1

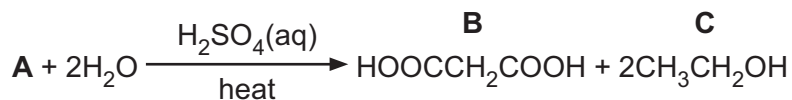
step 2

[2]

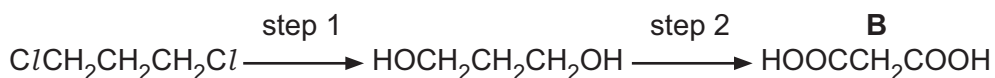
2. M/J 25/P21/Q5/d(i)

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

**Fig. 5.2**

(i) Identify the reagents and conditions required in steps 1 and 2.

step 1

step 2

[2]

3. M/J 25/P23/Q5/a(ii)

5 (a) **W** is a colourless liquid.

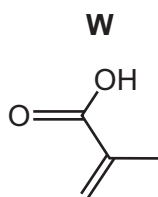


Fig. 5.1

(ii) Two different reagents are each added to separate samples of **W** as shown in Table 5.1.

Complete Table 5.1.

Table 5.1

reagent and conditions	observation when reagent is added to W	structural formula of organic product when reagent is added to W
solid Na		
Br ₂ in absence of ultraviolet light		

[4]

4. M/J 25/P24/Q4/d(iii)

4 The skeletal formula of **E** is shown in Fig. 4.1.

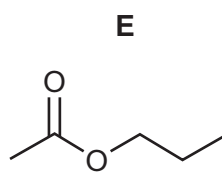


Fig. 4.1

(d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

(iii) CH_3OH is used to make CH_3CN in a two-step process, as shown in Fig. 4.2.

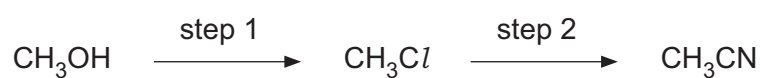


Fig. 4.2

State the reagents and conditions required for steps 1 and 2.

step 1

step 2

[3]

5. MAR 25/P22/Q4/a

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

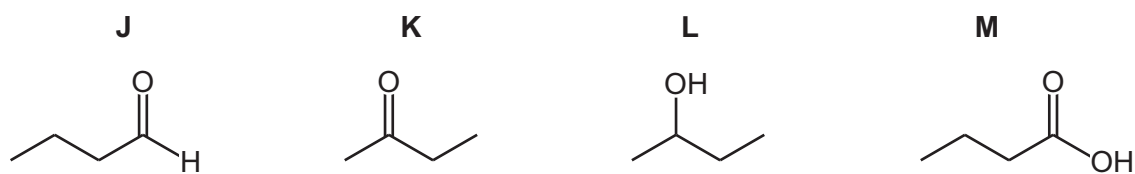


Fig. 4.1

- (a)** Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$		J and L
alkaline $\text{I}_2(\text{aq})$	yellow precipitate	
	orange precipitate	J and K
$\text{Na}(\text{s})$		

[5]

6. M/J 24/P22/Q4/c

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(c) Separate samples of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$, $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ and $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$ are tested with different reagents.

Complete Table 4.1. If no reaction occurs, write $\times$ in the relevant box.

Table 4.1

reagent added	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
$\text{Br}_2(\text{l})$ in the dark			
$\text{PCl}_5(\text{s})$			
$\text{AgNO}_3(\text{aq})$			

[3]

**7. O/N 23/P21/Q4/d**

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .



Fig. 4.1

(c) **C** and **D** both react with HBr .

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

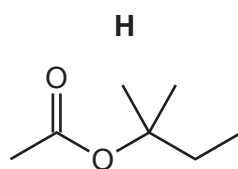
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

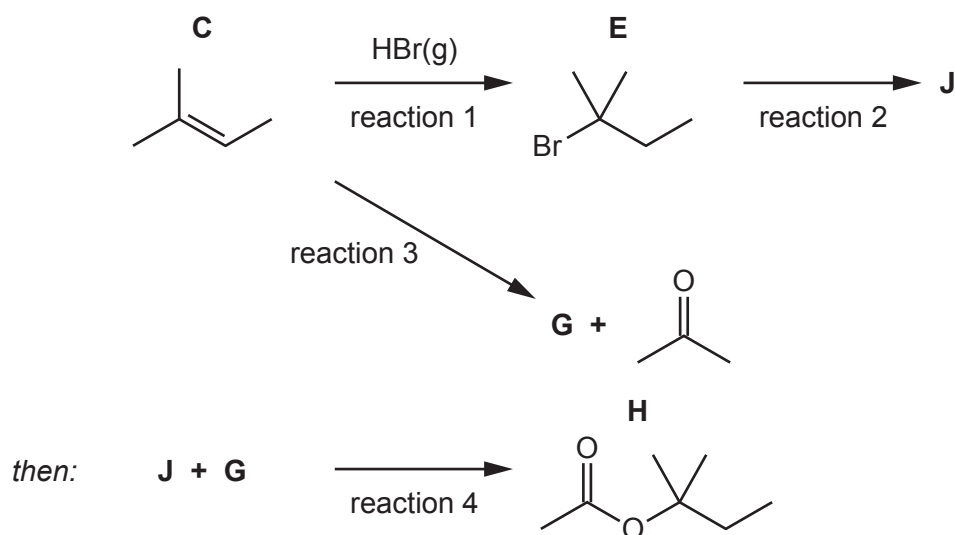
[3]

(d) **C** can be used to form **H**.

**Fig. 4.4**

One possible synthesis of **H** is shown in Fig. 4.5. Different portions of **C** are used in reactions 1 and 3. Some of the products are then combined to produce **H**.

Fig. 4.5 does not show any of the inorganic products of the reactions.

**Fig. 4.5**

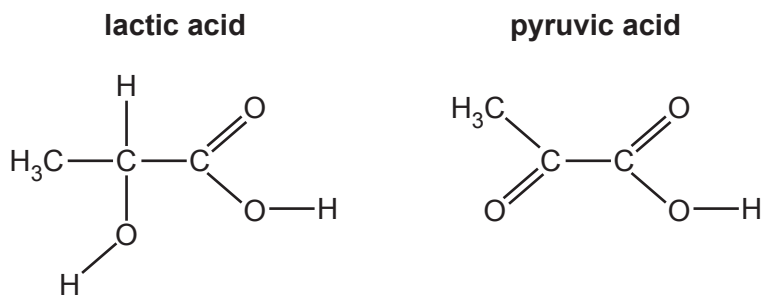
Complete Table 4.1 with the reagents and conditions required for each of the reactions shown in Fig. 4.5.

Table 4.1

	reagent and conditions
reaction 1 C $\rightarrow$ E	HBr(g)
reaction 2 E $\rightarrow$ J	
reaction 3 C $\rightarrow$ G +	
reaction 4 J + G $\rightarrow$ H	

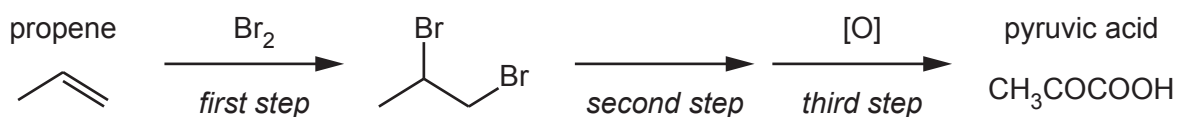
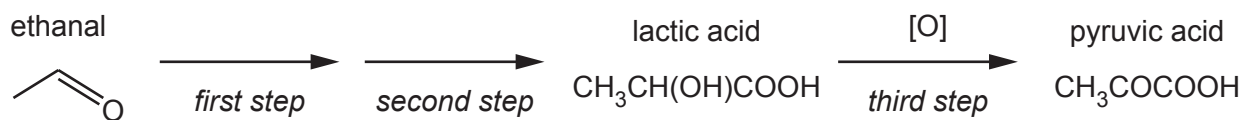
8. O/N 23/P22/Q4/b(iii)

- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (iii) Complete Table 4.1 to give details of the reagents and conditions used in each of the two syntheses shown in Fig. 4.3 and Fig. 4.4.

Table 4.1

		synthesis from propene (shown in Fig. 4.3)	synthesis from ethanal (shown in Fig. 4.4)
reagents and conditions used	<i>first step</i>	Br_2	
	<i>second step</i>		
	<i>third step</i>		

[4]

9. O/N 23/P23/Q4/d

- 4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10} .



Fig. 4.1

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(d) **C** can be used to form **H**.

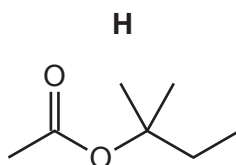
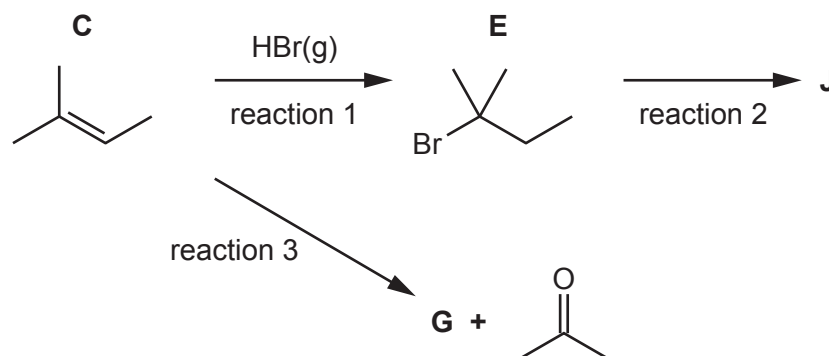


Fig. 4.4

One possible synthesis of **H** is shown in Fig. 4.5. Different portions of **C** are used in reactions 1 and 3. Some of the products are then combined to produce **H**.

Fig. 4.5 does not show any of the inorganic products of the reactions.



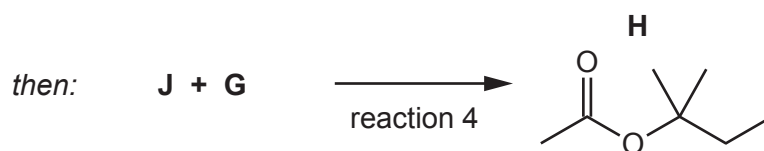


Fig. 4.5

Complete Table 4.1 with the reagents and conditions required for each of the reactions shown in Fig. 4.5.

Table 4.1

	reagent and conditions
reaction 1 $\text{C} \rightarrow \text{E}$	HBr(g)
reaction 2 $\text{E} \rightarrow \text{J}$	
reaction 3 $\text{C} \rightarrow \text{G} +$	
reaction 4 $\text{J} + \text{G} \rightarrow \text{H}$	

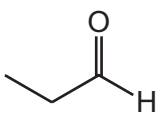
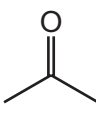
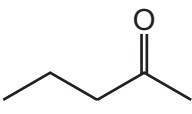
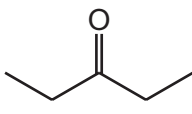
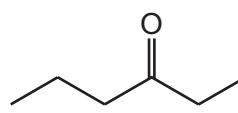
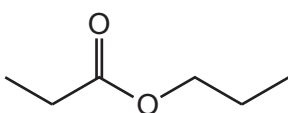
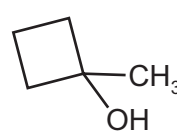
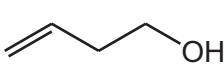
[3]

10. O/N 22/P22/Q3/a(i),d

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(a) Table 3.1 shows four pairs of organic compounds.

Table 3.1

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

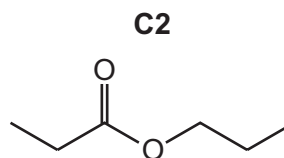
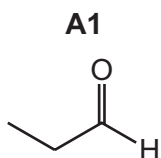
(i) Complete Table 3.1 to:

- identify a reagent which can distinguish between the compounds in each pair
- give the **positive** result of the chemical test **and** identify which compound shows this result.

Use a different reagent for each test.

[8]

(d) **C2** can be synthesised using **A1** as a single organic reactant.



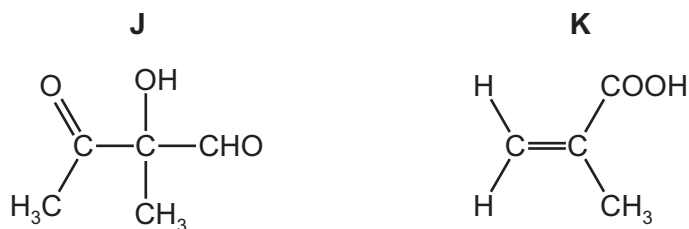
Devise a multi-step synthetic route to form **C2** from **A1**.

Identify relevant reagents and conditions, and state the organic products of each step.

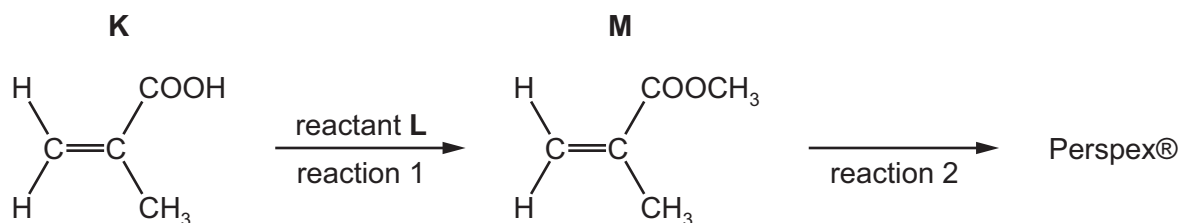
[3]

11. MAR 22/P22/Q4/b(iv)

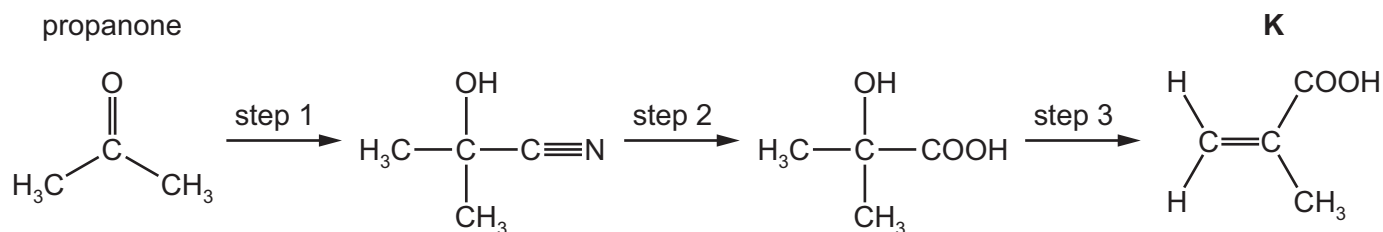
4 Compounds **J** and **K** are found in plant oils.

**Fig. 4.1**

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

**Fig. 4.2**

(iv) **K** can be made from propanone in the three-step synthesis shown in Fig. 4.3.

**Fig. 4.3**

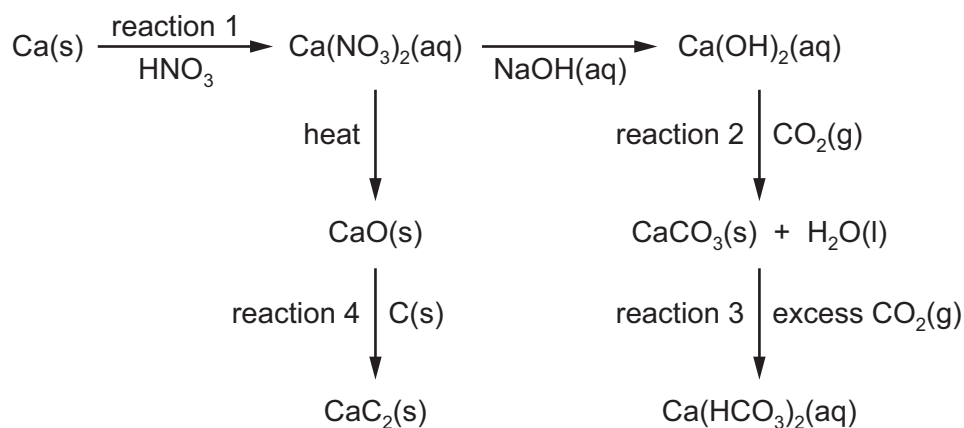
Complete Table 4.3 to identify the reagent(s) used and the type of reaction in each step.

Table 4.3

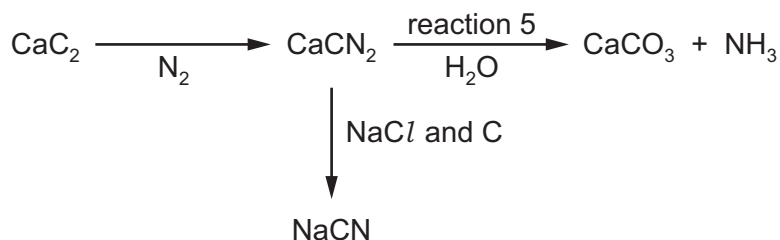
step	reagent(s)	type of reaction
1		
2		
3	Al_2O_3	

12. O/N 21/P21/Q2/d(ii)

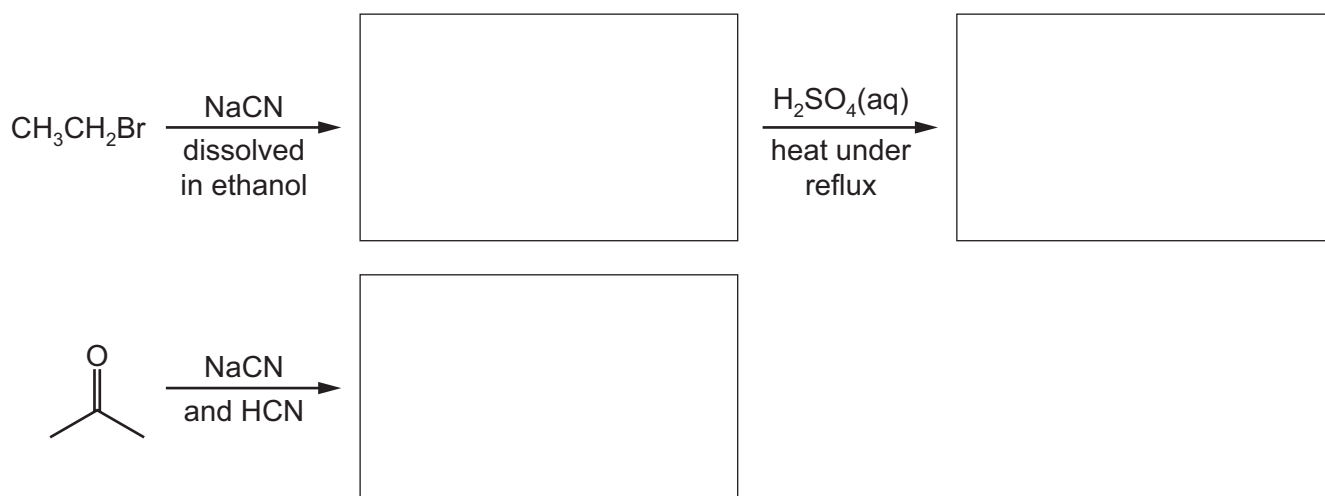
2 The reaction scheme shows some reactions of calcium.



(d) The flowchart shows some reactions of CaC_2 .



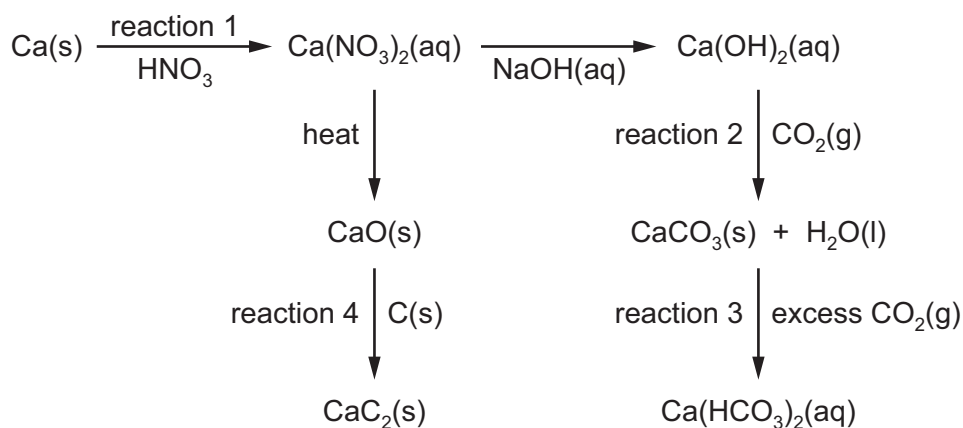
(ii) Draw the structure of the organic products formed in the following reactions.



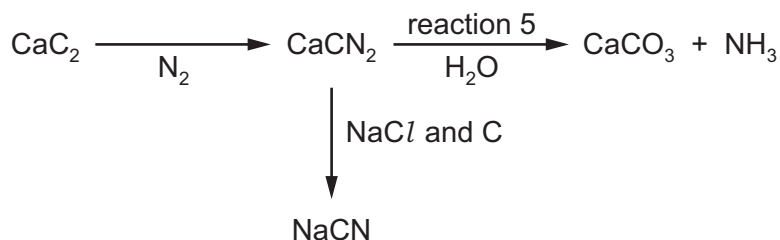
[3]

13. O/N 21/P23/Q2/d(ii)

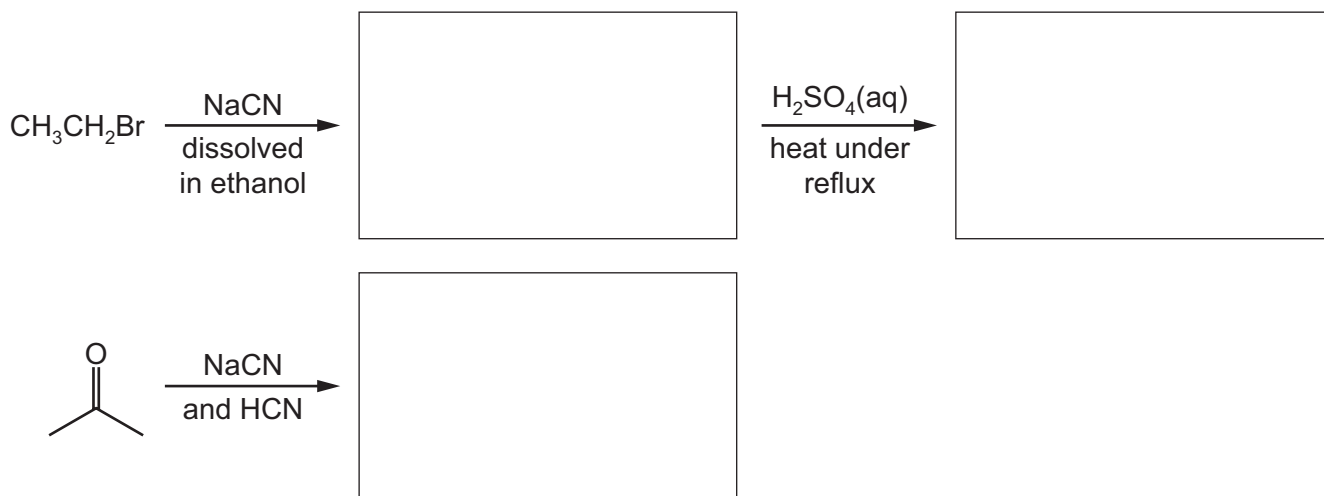
2 The reaction scheme shows some reactions of calcium.



(d) The flowchart shows some reactions of CaC_2 .



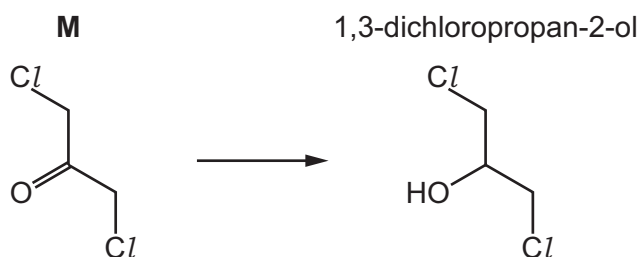
(ii) Draw the structure of the organic products formed in the following reactions.



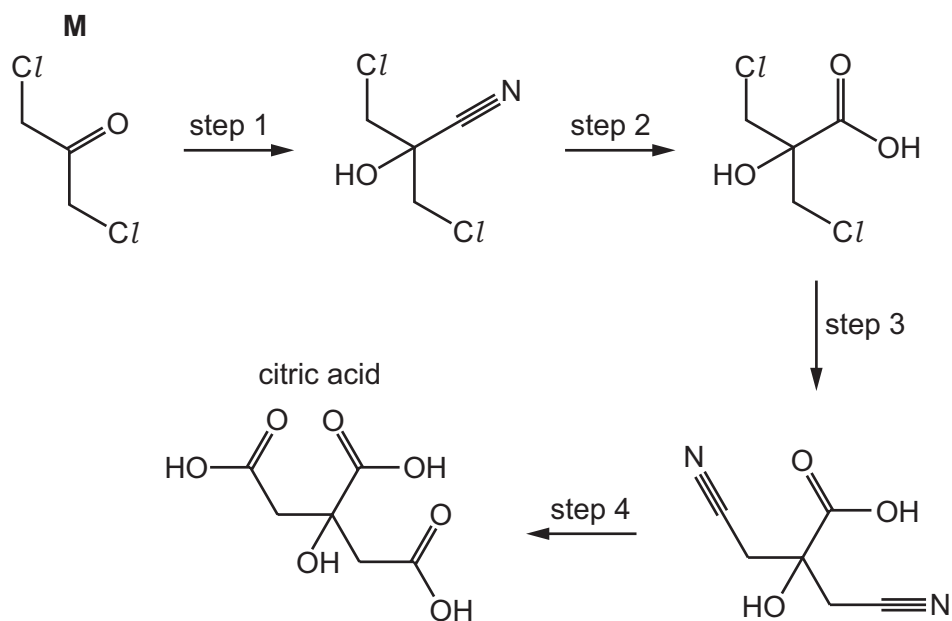
[3]

14. M/J 21/P23/Q4/c

4 (a) 1,3-dichloropropan-2-ol can be made by reacting **M**.



(c) Citric acid can be made from **M** in a four-step reaction.



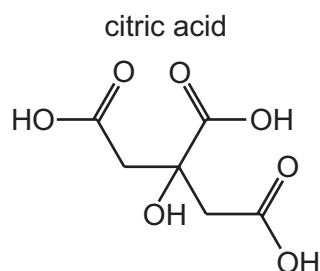
Complete the table for each step of the reaction sequence to identify:

- the reagents and conditions required
- the type of reaction.

step	reagent and conditions	type of reaction
1		
2	dilute sulfuric acid	
3		
4	dilute sulfuric acid	

15. M/J 21/P23/Q5/b(i)

5 Compound **X** contains the same functional groups as citric acid.



The table describes some of the similarities and differences between citric acid and compound **X**.

	citric acid	X
chiral centre	no	yes
reaction with Na	fizzing	fizzing
reaction with $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	remains orange	orange to green

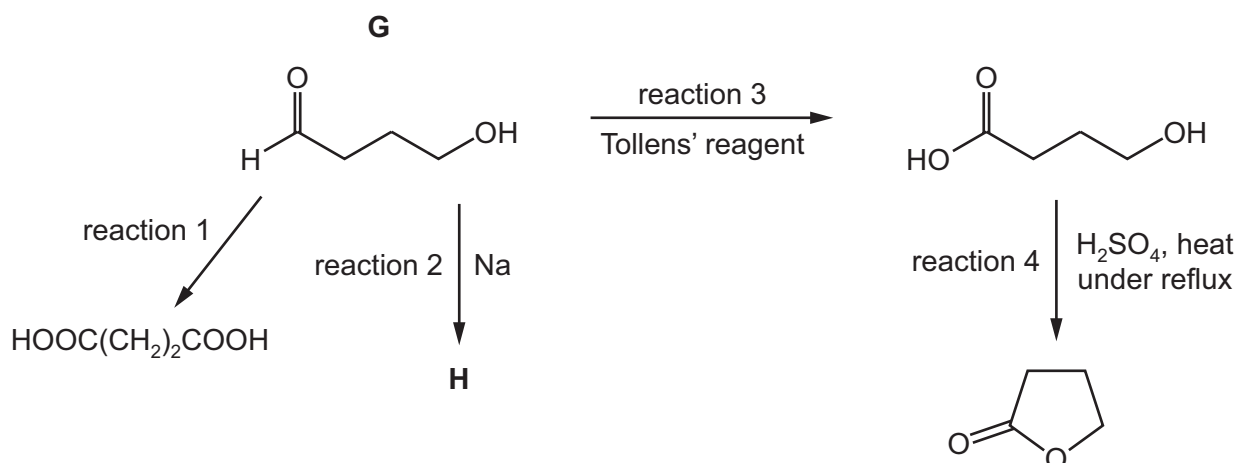
(b) (i) Use the information in the table to deduce the skeletal formula of **X**, $\text{C}_3\text{H}_6\text{O}_3$. Draw the skeletal formula of **X** in the box. Label the chiral centre of compound **X** with an asterisk (*).



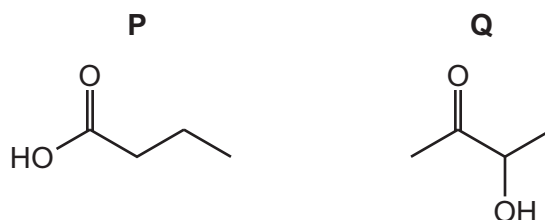
[2]

16. O/N 20/P22/Q4/c

4 Some reactions of compound **G** are shown.



(c) **P** and **Q** have the same molecular formula as **G**.



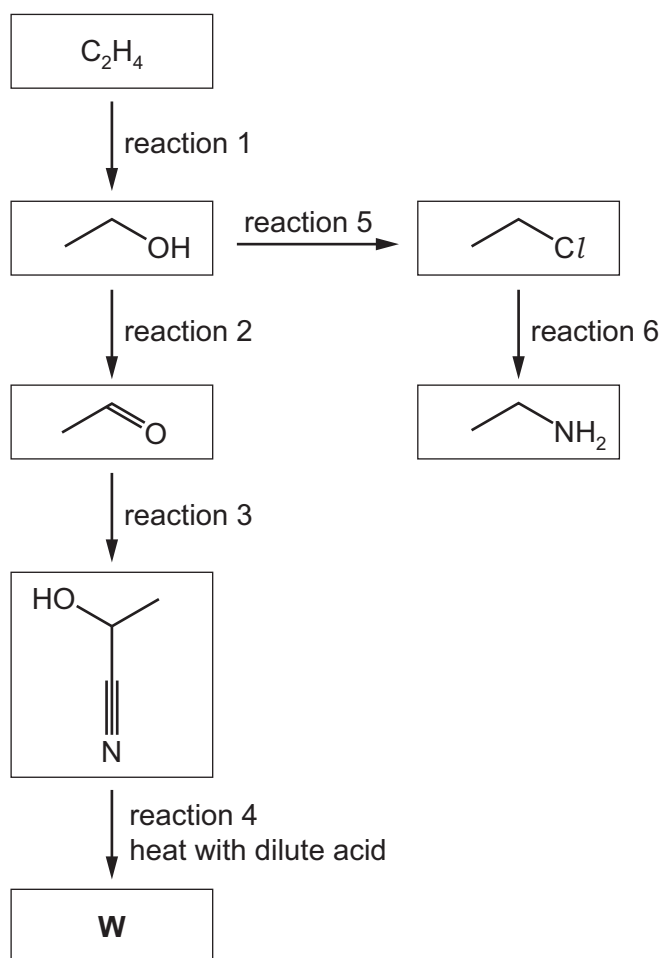
Complete the table with the expected observations for the reactions of **P** and **Q** with the named reagents.

reagent	result with P	result with Q
Br ₂ (aq)		
2,4-dinitrophenylhydrazine		
aqueous sodium carbonate		

[3]

17. M/J 20/P22/Q5/a

5 The reaction sequence shows how ethene, C_2H_4 , can be converted into other organic molecules.



(a) Complete the table to give

- the name of the reaction mechanisms of reactions 1 and 6
- the reagents and conditions required for reactions 1, 2 and 6.

reaction	name of mechanism	name of reagents and conditions
1		
2		
6		

18. O/N 19/P21/Q4/c(vi)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

(vi) Draw the skeletal formula of **Z**.

[3]

19. O/N 19/P23/Q4/c(vi)

4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(c) **X**, **Y** and **Z** are all isomers of $(\text{CH}_3)_3\text{CCHO}$.

A summary of some of the reactions and properties of **X**, **Y** and **Z** is shown in the table.

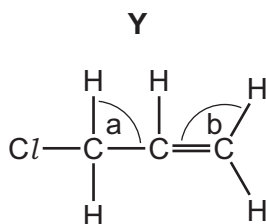
compound	observations with 2,4-DNPH	observations with Fehling's solution	principal absorptions in infra-red spectrum
X		no reaction	1715 cm^{-1}
Y		red precipitate	1730 cm^{-1}
Z	no reaction	no reaction	$3200\text{--}3600\text{ cm}^{-1}$ 1630 cm^{-1} 1050 cm^{-1}

(vi) Draw the skeletal formula of **Z**.

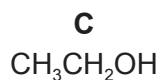
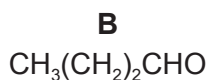
[3]

20. M/J 19/P21/Q4/d(ii)

4 The structure of compound **Y** is shown.



(d) Samples of organic compounds, **A**, **B**, **C** and **D**, are placed in unlabelled bottles.



(ii) **A–D** are reacted separately with the reagents given in the table.

Complete the table to:

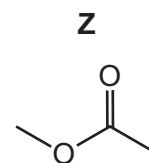
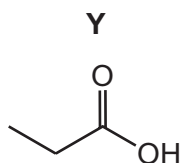
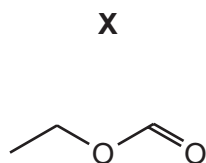
- identify which of the compounds, **A–D**, reacts with the reagents
- give an appropriate observation when a reaction occurs.

reagent	compounds identified	observation when a reaction occurs
Tollens' reagent		
alkaline solution of iodine		
sodium metal		

21. M/J 19/P22/Q5/c(ii)

5 Many naturally occurring esters are used as flavourings in food.

(c) Compounds **X**, **Y** and **Z** are shown. They all have the same molecular formula.



(ii) In three experiments, sodium is added to separate samples of **X**, **Y** and **Z**.

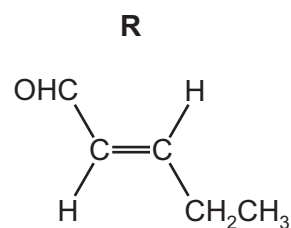
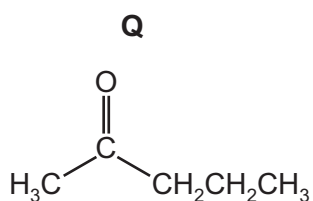
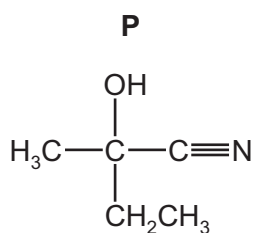
Complete the table to show the observations for each of these three experiments. Ignore any temperature changes which may occur.

experiment	observations
Na + X	
Na + Y	
Na + Z	

[2]

22. MAR 19/P22/Q3/a

3 **P**, **Q** and **R** all contain five carbon atoms.



A student carries out several tests to distinguish between **P**, **Q** and **R**.

(a) Complete the table, identifying any observations for the reaction of each reagent with **P**, **Q** and **R**.

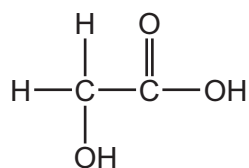
If no reaction occurs, write 'no reaction'.

reagent	observations with		
	P	Q	R
Na(s)			
2,4-DNPH	no reaction		
acidified K ₂ Cr ₂ O ₇ (aq)	no reaction		

[3]

23. O/N 18/P21/Q4/a

4 The structure of glycolic acid is shown.



glycolic acid

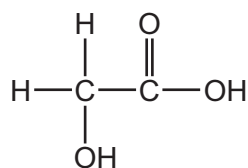
- (a)** Complete the table to show what you would **observe** when an aqueous solution of glycolic acid is added separately to each of the reagents. If a reaction occurs, state the functional group of glycolic acid that is responsible for the reaction.

reagent	observation with glycolic acid	does a reaction occur? ✓/✗	functional group
$\text{Na}_2\text{CO}_3(\text{aq})$			
2,4-DNPH			
acidified $\text{Cr}_2\text{O}_7^{2-}$			

[4]

24. O/N 18/P23/Q4/a

4 The structure of glycolic acid is shown.



glycolic acid

- (a) Complete the table to show what you would **observe** when an aqueous solution of glycolic acid is added separately to each of the reagents. If a reaction occurs, state the functional group of glycolic acid that is responsible for the reaction.

reagent	observation with glycolic acid	does a reaction occur? ✓/✗	functional group
$\text{Na}_2\text{CO}_3(\text{aq})$			
2,4-DNPH			
acidified $\text{Cr}_2\text{O}_7^{2-}$			

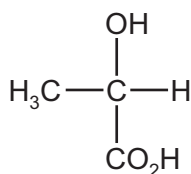
[4]

25. MAR 18/P22/Q3/d

also Unit 18 (1 mark), Unit 19 (1 mark), Unit 17 (1 mark), Unit 13 (1 mark); this unit 4 marks

3 Calcium and its compounds have a large variety of applications.

- (d) Calcium lactate is used in some medicines. It forms when lactic acid (2-hydroxypropanoic acid) reacts with calcium carbonate.

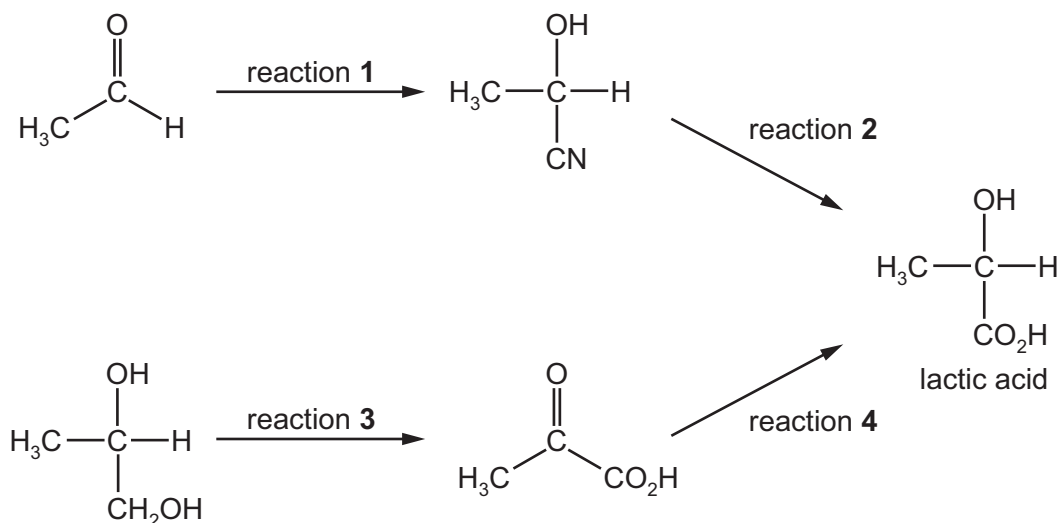


lactic acid

- (i) Identify the **two** other products of the reaction of lactic acid with calcium carbonate.

[1]

Two possible methods of making lactic acid are shown.



(ii) State suitable reagents and conditions for reactions 1 and 3.

reaction	reagents and conditions
1	
3	

[4]

(iii) Name the type of reaction that occurs in reaction 2.

[1]

(iv) Reaction 4 uses NaBH_4 .

Identify the role of NaBH_4 in this reaction.

[1]

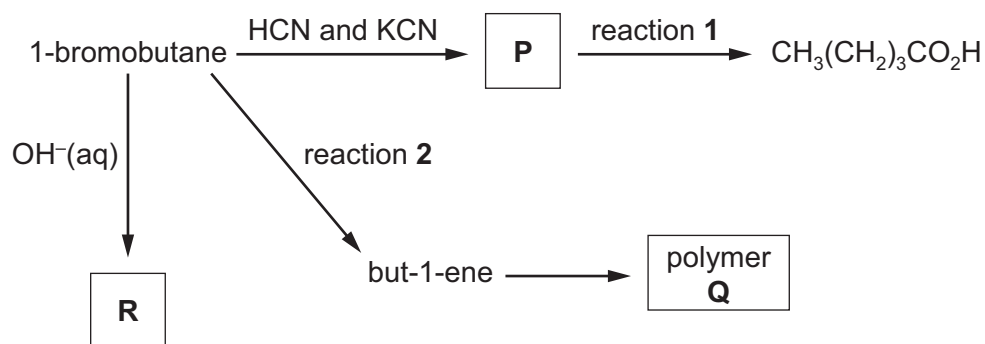
(v) Lactic acid has a chiral centre.

State what is meant by the term *chiral centre*.

[1]

26. MAR 17/P22/Q3/a(ii)

3 (a) A series of reactions starting from 1-bromobutane is shown.



(ii) Identify the reagent(s) and conditions for reactions **1** and **2**.

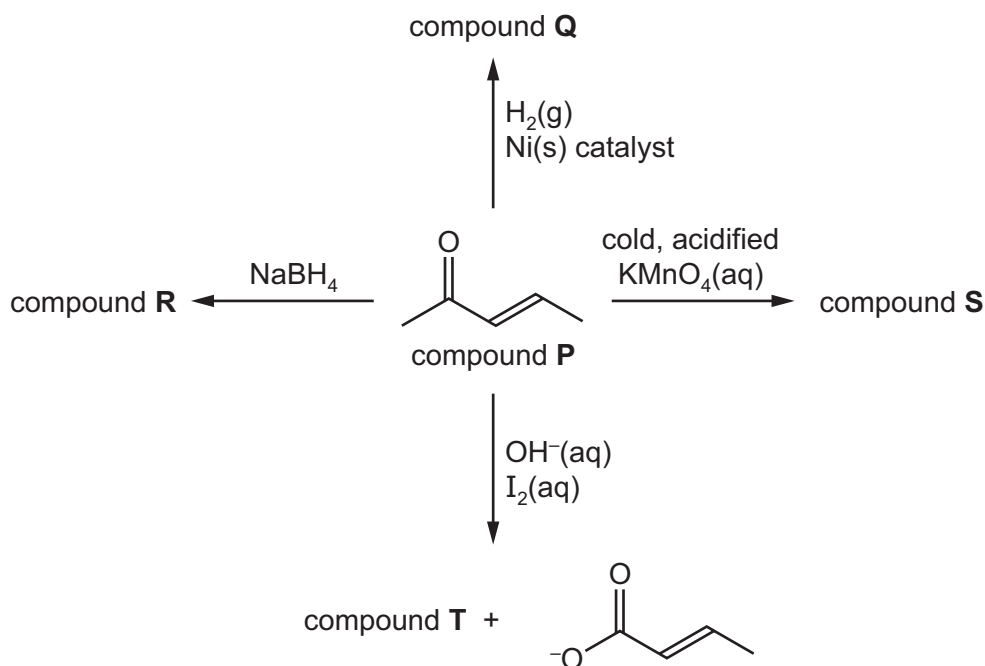
reaction **1**

reaction **2**

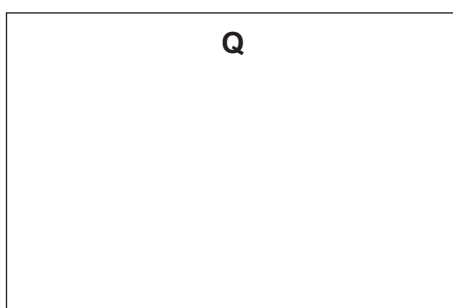
[2]

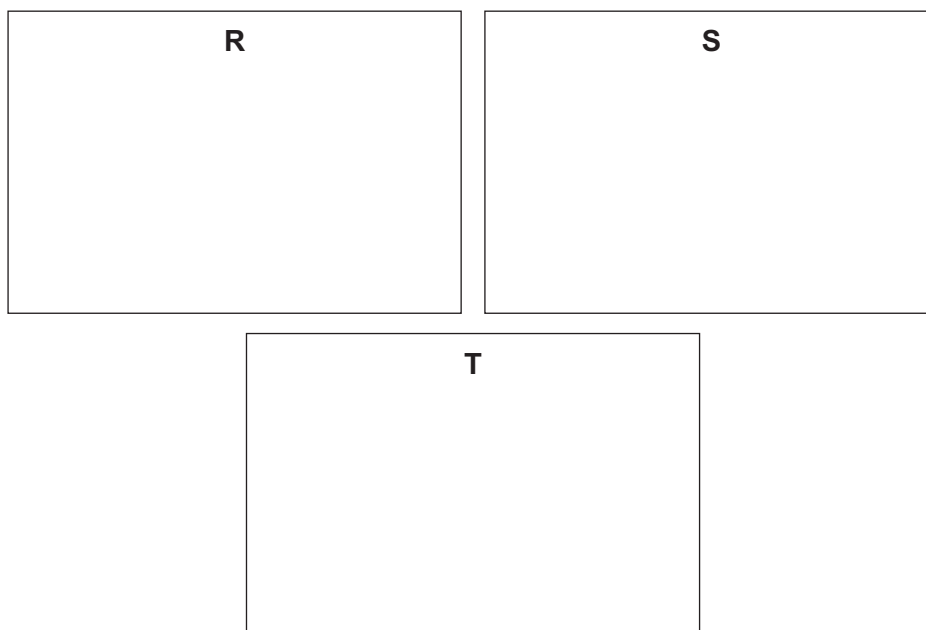
27. MAR 16/P22/Q5/a(i),b

5 Some reactions of compound **P**, $\text{C}_5\text{H}_8\text{O}$, are shown.



(a) (i) Give the structures for organic compounds **Q**, **R**, **S** and **T**.



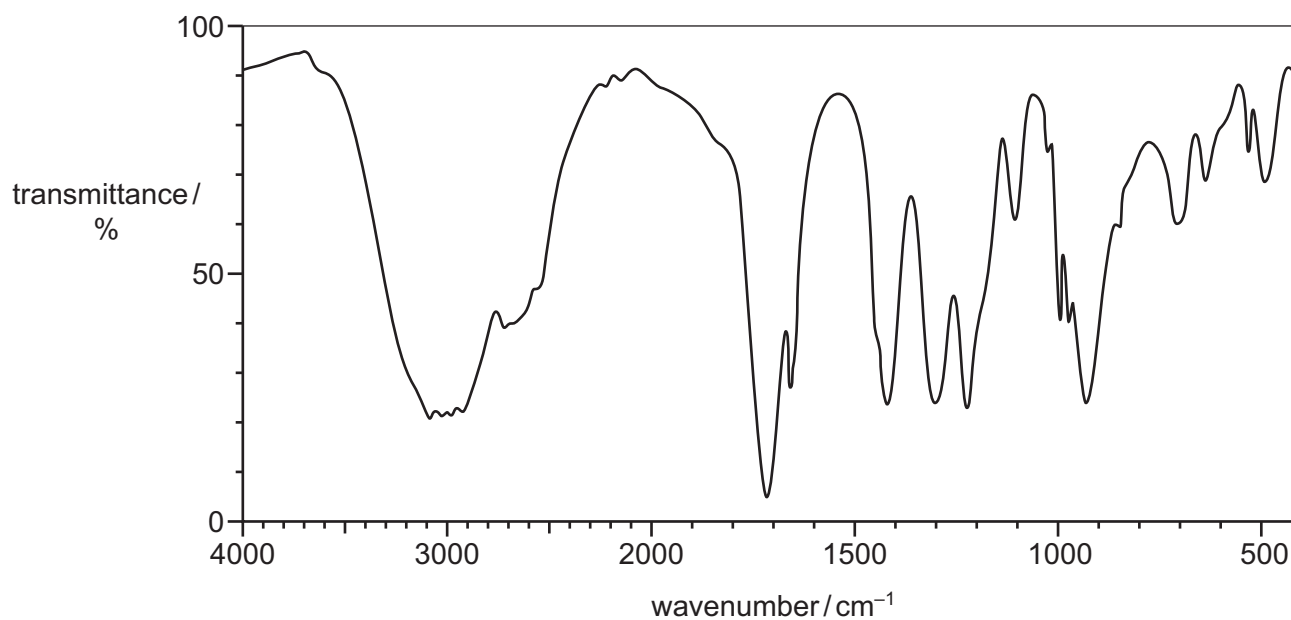


[4]

(b) Compound **U** contains a chiral centre and has the same molecular formula as compound **P**, C_5H_8O .

- Compound **U** readily decolourises a sample of bromine water.
- Compound **U** does not show cis-trans isomerism.
- When compound **U** is heated under reflux in the presence of excess acidified potassium dichromate(VI), the organic product gives the infra-red spectrum shown.

infra-red spectrum of **product**



Use the information given to suggest a structure for compound **U**.
Explain your answer.

[4]

Answers Section

1. M/J 26/P24/Q4/d(i)

4(d)(i)	M1 (step 1) Cl_2 AND ultraviolet (light) / UV (light) M2 (step 2) KCN in ethanol AND heat (under reflux)	2	M2 A ethanolic KCN AND heat alcohol = (alc) A NaCN in ethanol AND heat R HCN R KCN in (aq) ethanol R ethanoic I warm high temp = heat = reflux = $> 60^\circ\text{C}$
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2. M/J 25/P21/Q5/d(i)

5(d)(i)	M1 NaOH(aq) AND heat M2 acidified AND potassium dichromate(VI) AND heat under reflux	2
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3. M/J 25/P23/Q5/a(ii)

5(a)(ii)	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%;">reagent</th><th style="width: 35%;">observation when reagent is added to W</th><th style="width: 50%;">structural formula of organic product when reagent is added to W</th></tr> </thead> <tbody> <tr> <td>Na</td><td>fizz / bubbles etc / OR solid / Na disappears / dissolves</td><td>$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$</td></tr> <tr> <td>$\text{Br}_2$ in the dark</td><td>brown to colourless / decolourise</td><td>$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$</td></tr> </tbody> </table>	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$	Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$	4
reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W									
Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$									
Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$									

4. M/J 25/P24/Q4/d(iii)

4(d)(iii)	M1 step 1 PCl_3 and heat OR $\text{HCl}(\text{g})$ OR PCl_5 OR SOCl_2 OR NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 M2 step 2 reagent KCN M3 conditions for step 2 dissolved in ethanol AND heat	3
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5. MAR 25/P22/Q4/a

4(a)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tbody> <tr> <td style="width: 30%;">acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$</td><td style="width: 30%;">(solution) turns (from orange to) green</td><td style="width: 40%; text-align: center;">J and L</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>yellow precipitate forms</td><td style="text-align: center;">K and L</td></tr> <tr> <td>2,4-dinitrophenylhydrazine / 2,4-DNPH</td><td>orange precipitate forms</td><td style="text-align: center;">J and K</td></tr> <tr> <td>Na(s)</td><td>effervescence</td><td style="text-align: center;">L and M</td></tr> </tbody> </table>	acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L	alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L	2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K	Na(s)	effervescence	L and M	5
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L												
alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L												
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K												
Na(s)	effervescence	L and M												

6. M/J 24/P22/Q4/c

4(c)

Chemical test	observation with 2-bromooctane	observation with octan-2-ol	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
Br_2 in the dark	✗		orange / brown to colourless
PCl_5		steamy fumes	✗
$\text{AgNO}_3(\text{aq})$	cream precipitate OR off white precipitate	✗	

6 correct answers = 3 marks
4 or 5 correct answers = 2 marks
2 or 3 correct answers = 1 mark
0 or 1 correct answer = 0 marks

3

7. O/N 23/P21/Q4/d

4(d)	(reaction 2) $\text{NaOH}(\text{aq})$	1
	(reaction 3) hot acidified concentrated KMnO_4	1
	(reaction 4) <u>conc</u> H_2SO_4 catalyst	1

8. O/N 23/P22/Q4/b(iii)

4(b)(iii)	$\text{HCN}\&\text{KCN}$ OR NaCN with H_2SO_4	1
	$\text{NaOH}(\text{aq})$ (1) dilute / aq(ueous) HCl / H_2SO_4 (1)	2
	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	1

9. O/N 23/P23/Q4/d

4(d)	(reaction 2) $\text{NaOH}(\text{aq})$	1
	(reaction 3) hot acidified concentrated KMnO_4	1
	(reaction 4) <u>conc</u> H_2SO_4 catalyst	1

10. O/N 22/P22/Q3/a(i),d

3(a)(i)	M1 Tollens' OR Fehling's OR $\text{K}_2\text{Cr}_2\text{O}_7$ OR KMnO_4 OR alkaline $\text{I}_2(\text{aq})$ with	M2 A1 gives silver mirror OR brick-red ppt OR (orange solution) turns green OR (purple solution) turns colourless A2 gives yellow ppt	2
	M3 alkaline $\text{I}_2(\text{aq})$	M4 B1 gives yellow ppt	2
	M5 2,4-DNPH	M6 C1 gives red/orange/yellow ppt	2
	M7 $\text{Br}_2(\text{aq})$	M8 D2 turns it (orange to) colourless	2
3(d)	M1 one portion of A1 : NaBH_4 AND $\rightarrow$ propan-1-ol / $\text{CH}_3(\text{CH}_2)_2\text{OH}$		1
	M2 second portion of A1 : acidified $\text{K}_2\text{Cr}_2\text{O}_7$ AND $\rightarrow$ propanoic acid / $\text{C}_2\text{H}_5\text{CO}_2\text{H}$		1
	M3 two products: <u>conc</u> (entrated mineral) acid / H^+ / H_2SO_4 (catalyst)		1

11. MAR 22/P22/Q4/b(iv)

4(b)(iv)	step 1 step 2 step 3	KCN / HCN OR NaCN / H ₂ SO ₄ [1] H ⁺ / H ₂ SO ₄ (aq) [1]	addition [1] hydrolysis / substitution [1] elimination / dehydration [1]	5
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12. O/N 21/P21/Q2/d(ii)

2(d)(ii)		CH ₃ CH ₂ CN	CH ₃ CH ₂ CO ₂ H	3
		CH ₃ C(OH)(CN)CH ₃		

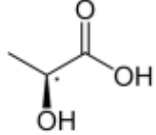
13. O/N 21/P23/Q2/d(ii)

2(d)(ii)		CH ₃ CH ₂ CN	CH ₃ CH ₂ CO ₂ H	3
		CH ₃ C(OH)(CN)CH ₃		

14. M/J 21/P23/Q4/c

4(c)	<table> <tr> <th>step</th><th>reagent and conditions</th><th>type of rxn</th></tr> <tr> <td>1</td><td>M1 HCN and (trace) KCN / NaCN</td><td>M3 Addition</td></tr> <tr> <td>2</td><td>dilute sulfuric acid</td><td>Hydrolysis</td></tr> <tr> <td>3</td><td>M2 KCN/NaCN in ethanol / alcohol</td><td>M4 Substitution</td></tr> <tr> <td>4</td><td>dilute sulfuric acid</td><td>Hydrolysis</td></tr> </table> M5 type of reaction for step 2 AND step 4	step	reagent and conditions	type of rxn	1	M1 HCN and (trace) KCN / NaCN	M3 Addition	2	dilute sulfuric acid	Hydrolysis	3	M2 KCN/NaCN in ethanol / alcohol	M4 Substitution	4	dilute sulfuric acid	Hydrolysis	5
step	reagent and conditions	type of rxn															
1	M1 HCN and (trace) KCN / NaCN	M3 Addition															
2	dilute sulfuric acid	Hydrolysis															
3	M2 KCN/NaCN in ethanol / alcohol	M4 Substitution															
4	dilute sulfuric acid	Hydrolysis															

15. M/J 21/P23/Q5/b(i)

5(b)(i)	M1 skeletal formula of 2-hydroxypropanoic acid / similar chiral isomer that meet the criteria in table 	1
	M2 chiral carbon marked with *	1

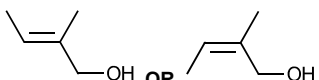
16. O/N 20/P22/Q4/c

4(c)	<table> <tr> <th>reagent</th><th>result with P</th><th>result with Q</th></tr> <tr> <td>Br₂(aq)</td><td>no change / stays orange</td><td>no change / stays orange</td></tr> <tr> <td>2,4-DNPH</td><td>no change</td><td>orange ppt</td></tr> <tr> <td>Na₂CO₃</td><td>effervescence</td><td>no change</td></tr> </table> Award one mark for every two correct observations.	reagent	result with P	result with Q	Br ₂ (aq)	no change / stays orange	no change / stays orange	2,4-DNPH	no change	orange ppt	Na ₂ CO ₃	effervescence	no change	3
reagent	result with P	result with Q												
Br ₂ (aq)	no change / stays orange	no change / stays orange												
2,4-DNPH	no change	orange ppt												
Na ₂ CO ₃	effervescence	no change												

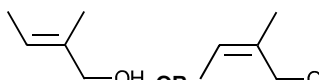
17. M/J 20/P22/Q5/a

5(a)	Rxn.	name of mechanism	Name of reagents and conditions	6
	1	M1 electrophilic addition	M2 steam AND concentrated phosphoric acid (catalyst)	
	2		M3 & M4 Two marks for name of reagent and both conditions. One mark for name of reagent and one conditions acidified potassium dichromate ((VI)) condition 1 warm condition 2 distil NOT reflux	
	6	M5 nucleophilic substitution	M6 ammonia (alcoholic) AND heat in a sealed tube / heat under pressure	

18. O/N 19/P21/Q4/c(vi)

4(c)(vi)	 M1: skeletal alkene group AND C5 structure M2: one alcohol group M3: branched chain AND capable of geometrical isomerism	3
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19. O/N 19/P23/Q4/c(vi)

4(c)(vi)	 M1: skeletal alkene group AND C5 structure M2: one alcohol group M3: branched chain AND capable of geometrical isomerism	3
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20. M/J 19/P21/Q4/d(ii)

4(d)(ii)			Compound(s)	Observation	2
		Reaction with Tollens' reagent	B ✓	silver mirror OR grey / black / brown / silver precipitate ✓	
			Compound(s)	Observation	3
		Reaction with alkaline aq. iodine	A ✓ and C ✓	(Pale) yellow precipitate /solid ✓	
			Compound(s)	Observation	3
		Reaction with sodium metal	C ✓ and D ✓	Effervescence / sodium/solid disappears ✓	

21. M/J 19/P22/Q5/c(ii)

5(c)(ii)	X and Z – no reaction / no (visible) change	1
	Y – effervesces	1

22. MAR 19/P22/Q3/a

3(e)			P	Q	R	3
		Na(s)	effervescence	no reaction	no reaction	
		2,4-DNPH	no reaction	orange ppt	orange ppt	
		acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	no reaction	no reaction	(turns) green	

23. O/N 18/P21/Q4/a

4(a)

reagent	observation with glycolic acid	does a reaction occur? ✓ / x	functional group
$\text{Na}_2\text{CO}_3(\text{aq})$	effervescence / fizzing / bubbling	✓	COOH / carboxylic acid
2,4–DNPH	no visible reaction owtte	x	(no group required)
acidified $\text{Cr}_2\text{O}_7^{2-}$	orange to green	✓	$-\text{OH}$ / alcohol

1 mark for each in column 2 (obs)

1 mark for COOH **and** OH

4

24. O/N 18/P23/Q4/a

4(a)

reagent	observation with glycolic acid	does a reaction occur? ✓ / x	functional group
Na ₂ CO ₃ (aq)	effervescence / fizzing / bubbling	✓	COOH / carboxylic acid
2,4–DNPH	no visible reaction owtte	x	(no group required)
acidified Cr ₂ O ₇ ²⁻	orange to green	✓	–OH / alcohol

1 mark for each in column 2 (obs)

1 mark for COOH **and** OH

4

25. MAR 18/P22/Q3/d

3(d)(i)	carbon dioxide AND water		1
3(d)(ii)	reaction	reagent(s) and condition(s)	4
	1	HCN ✓ NaCN ✓	
	3	• K₂Cr₂O₇ • H₂SO₄ / acid / H⁺ • (heat under) reflux	
3(d)(iii)	hydrolysis		1

3(d)(iv)	reducing agent	1
3(d)(v)	has a carbon / C / atom attached to four different groups / atoms / chains OR has no plane / line of symmetry / has non-superimposable images	1

26. MAR 17/P22/Q3/a(ii)

3(a)(ii)	reaction 1 = HCl(aq)	1
	reaction 2 = (conc.) NaOH/KOH AND ethanol	1

27. MAR 16/P22/Q5/a(i),b

5 (a) (i)	Q S R T 	[1] [1] [1] [1]	[4]
(b)	<i>This question was discounted.</i> M1 = decolourises bromine / $1500\text{--}1600\text{ cm}^{-1}$ = alkene M2 = absorption at 1700 cm^{-1} is C=O AND (very) broad absorption at $2500\text{--}3000\text{ cm}^{-1}$ is O—H = carboxylic acid M3 = no cis-trans so terminal alkene OR chiral so contains a carbon atom with 4 different groups attached M4 = U is	[1] [1] [1] [1]	[4]